

Sequential Models and RNNs

DATA 443/643 · Week 2, note 2.1

After Stanford CS224N W26 L4: any-length input, shared weights, then an RNN-LM.

Lecture plan

- 1 Why a fixed window is not enough
- 2 The cell, then the unroll
- 3 Many-to-one versus many-to-many
- 4 An RNN language model is next-token NLL
- 5 A two-step numeric table

Key goal: leave with a shared cell and a next-token loss that sits on a hidden state.

First time: what / why / formula

- What / why. Hidden state h_t is a function of h_{t-1} and the new token. An RNN-LM predicts the next token from h_t . A fixed window (n-gram / bag) cannot count “the verb agrees with the noun 40 tokens back” without growing width.
- Architecture. Shared cell $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$, then softmax on h_t . Unroll in time for training (BPTT).
- Formula. $p(w_t | w_{<t}) = \text{softmax}(W_o h_t)_w$.
- Tradeoffs. + Variable length, parameter sharing. – Sequential (hard to parallelize), vanishing/exploding gradients (next note).

How. Read left to right. The same W at every step. Lab 2 copies the first bit at the end.

A window cannot grow with the sentence

n-gram / fixed MLP

Last k words only. Raise k and the table is sparse, or the concat layer gets a new weight per position.

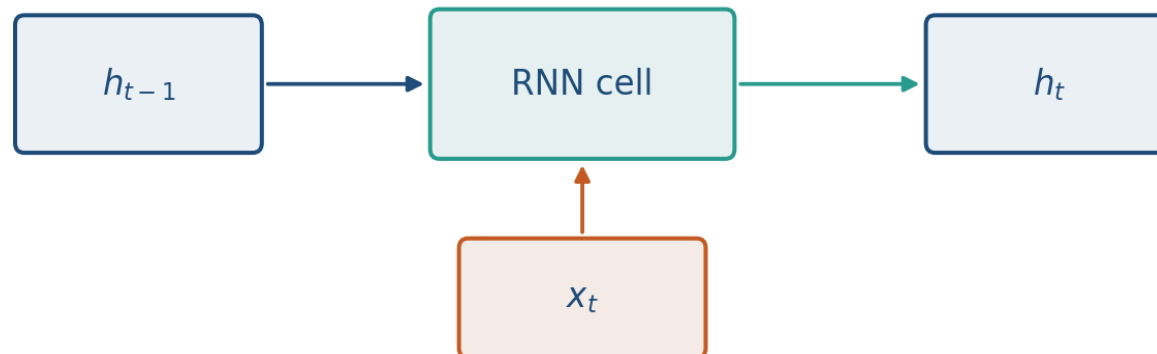
Recurrent cell

Any length. Same W_h at every step. In principle the first token can still sit in h_T .

The cell

- Each step: x_t and h_{t-1} become h_t .
- The same weights at every token.
- Reset $h_0 = 0$ between unrelated documents.

Same weights at every step: input plus previous state



Vanilla recurrence

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$

Copies of W_h , not a new matrix per step.

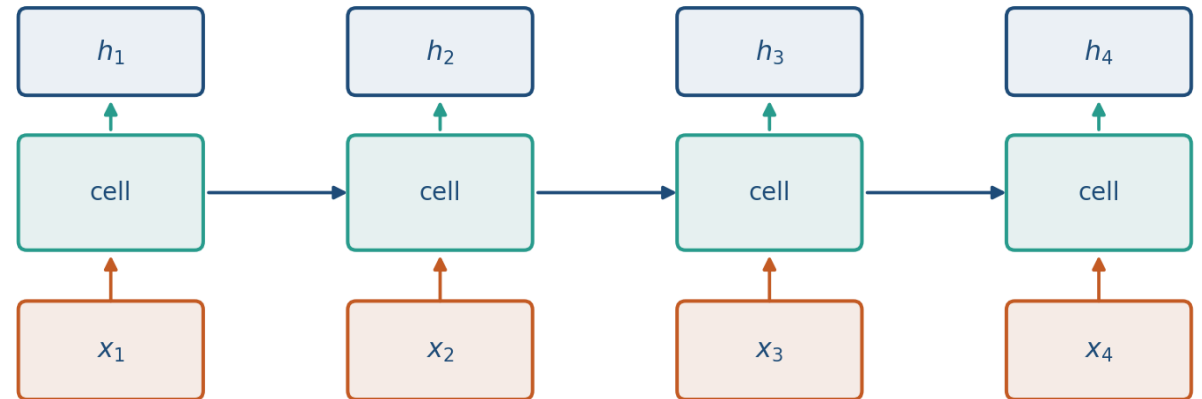
$\tanh$ is the usual vanilla default.

That overwrite is why long-range gradients die in 2.2.

Unrolled in time

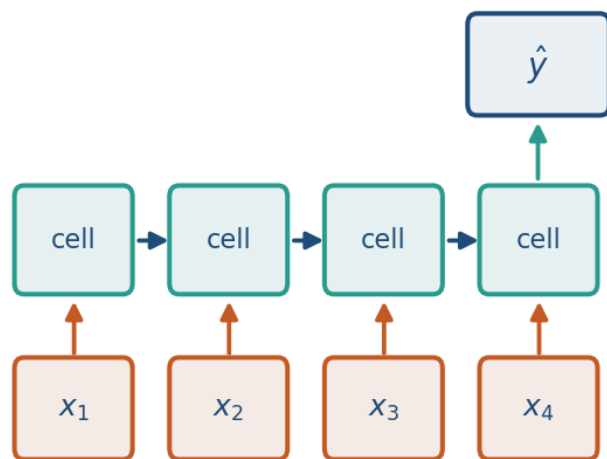
- The loop is a chain. That chain is the long-range path.
- A CNN is parallel with a fixed window.
- An RNN is sequential with the whole past.

Unrolled in time. The chain is why long-range gradients struggle.

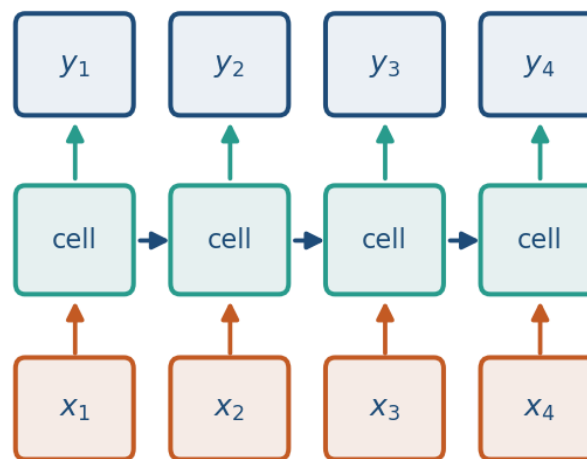


The same cell, different wiring

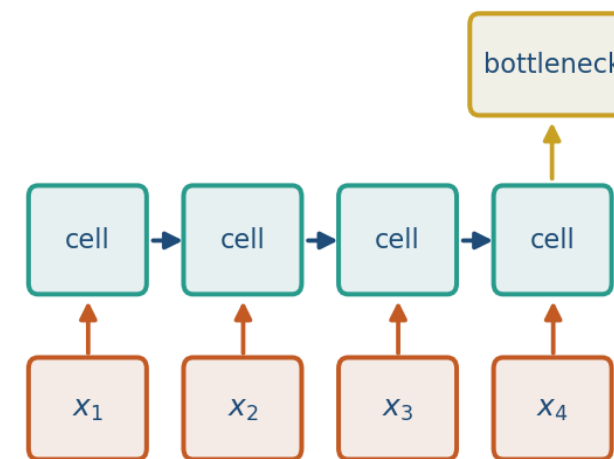
Many-to-one
(sentiment)



Many-to-many
(tagging / LM)



Encoder then decode
(leave for Week 3)

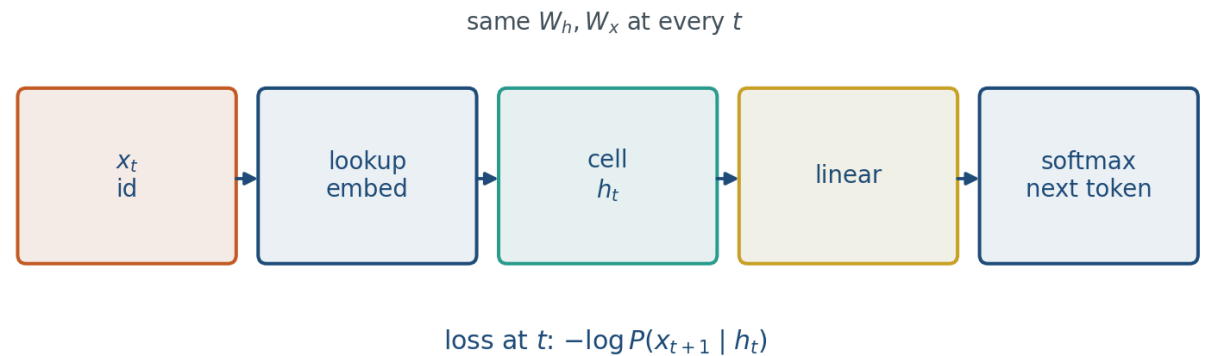


Lab 2 is many-to-one. Language modeling is many-to-many. Encoder-decoder waits for Week 3.

An RNN language model

- Softmax over the vocabulary from h_t .
- Loss at t : $-\log P(x_{t+1} | h_t)$.
- Teacher forcing: the true previous token, not a sample.

An RNN language model is next-token softmax from the hidden state



Two steps, same $(W_h=0.5I)$

$W_h = 0.5I$, $W_x = I$, $h_0 = 0$. Token 1 shrinks in h_2 .

- $h_0 = 0$, $x_1 = [1, 0]$, $x_2 = [0, 1]$.
- $h_1 \approx [0.76, 0]$. Then $h_2 \approx [0.36, 0.76]$.
- Token 1 shrinks in h_2 . That is the teaser for 2.2.

step	x_t	$W_h h_{t-1} + W_x x_t$	$h_t = \tanh(\cdot)$
$t = 1$	$[1, 0]$	$[1, 0]$	$[0.76, 0]$
$t = 2$	$[0, 1]$	$[0.38, 1]$	$[0.36, 0.76]$

What to take from this lecture

Share it

One W_h along the sentence. Reset between documents.

Score it

Next-token NLL from h_t . Same loss as Week 1, new state.

Watch it

If the first token has already shrunk, Lab 2 will fail at long T.